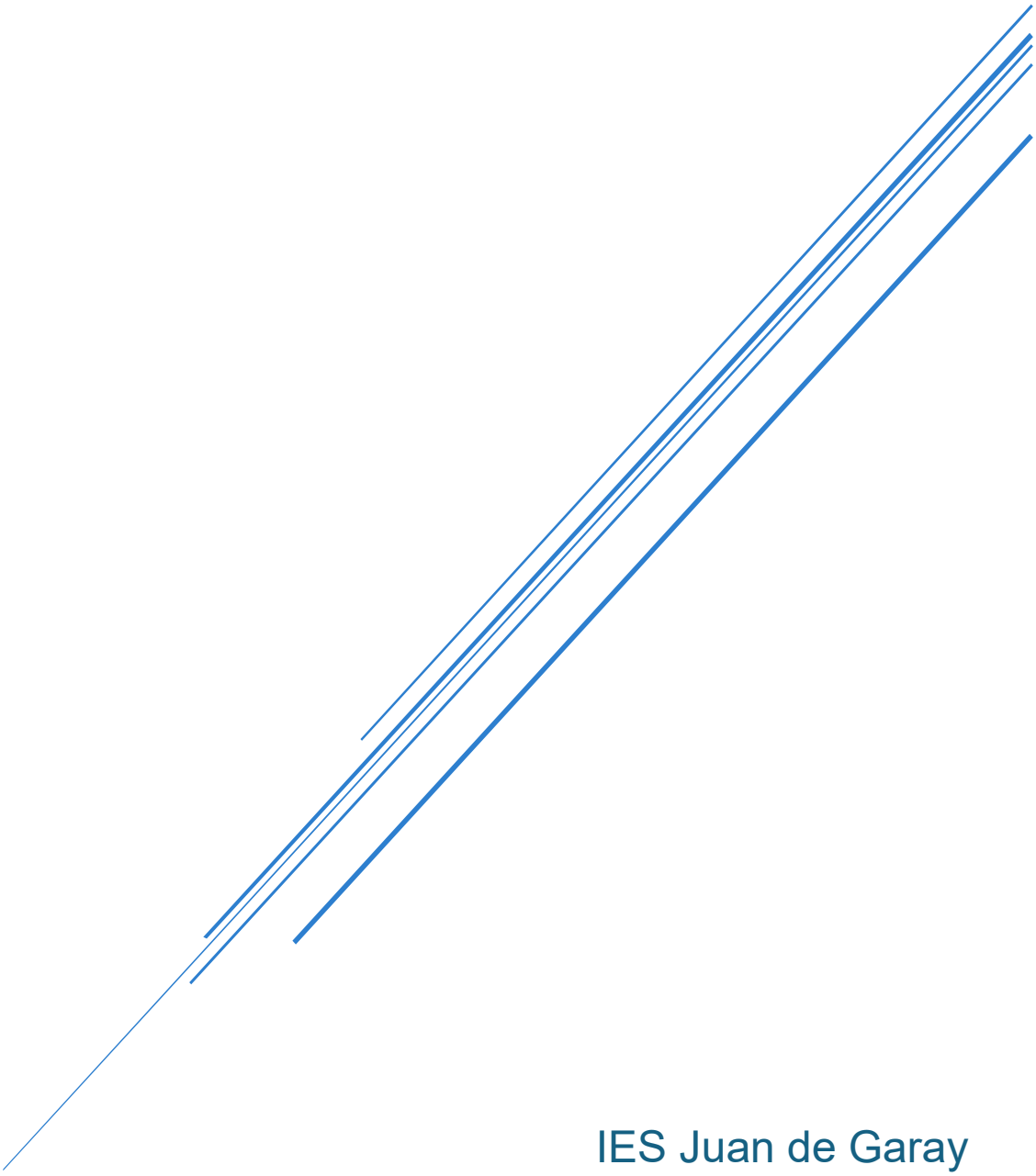
TAREA 1.2 – WORDPRESS EN DOS CAPAS

Roberto Ripoll Cones

07/11/2026



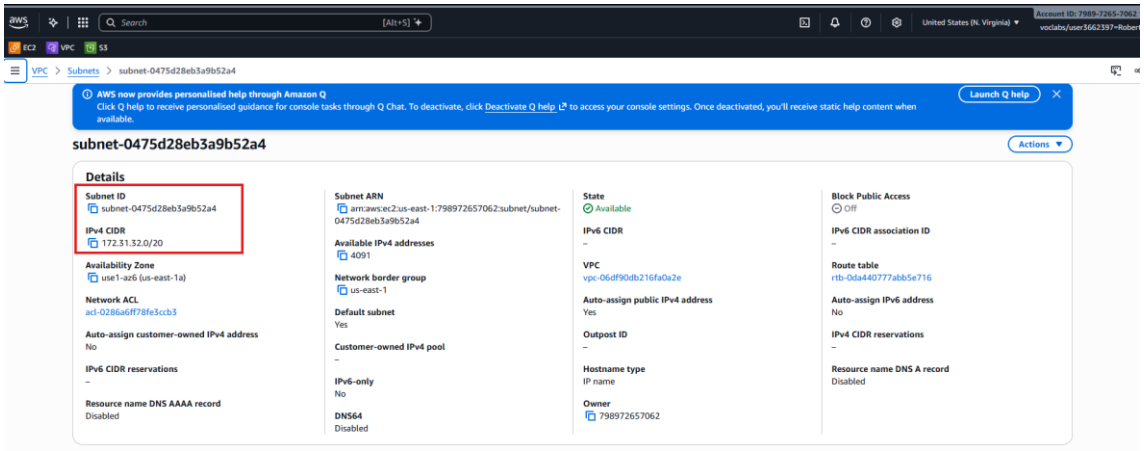
IES Juan de Garay

CE CLOUD – Administración de recursos de computación en la nube

Contenido

1. Preparación 2
2. Grupos de seguridad 2
3. Instancias e IP elásticas 4
4. Acceso SSH con host de salto 7
5. Capa de datos con MariaDB 8
6. Capa web con Apache, PHP y Wordpress 9
7. Primera entrada y comprobaciones 10
8. Base de datos en una subred privada 11
9. Para pensar 16
   9.1. ¿Qué ventaja tiene usar como origen el grupo wp-web-xx-sg en lugar de la IP privada de wp-web-xx? 16
   9.2. wp-bd-xx tiene IP pública, pero nadie puede conectarse a ella desde Internet. ¿Qué elemento lo impide? ¿Por qué no basta con quitarle la IP pública para que siga funcionando todo igual? 16

1. Preparación

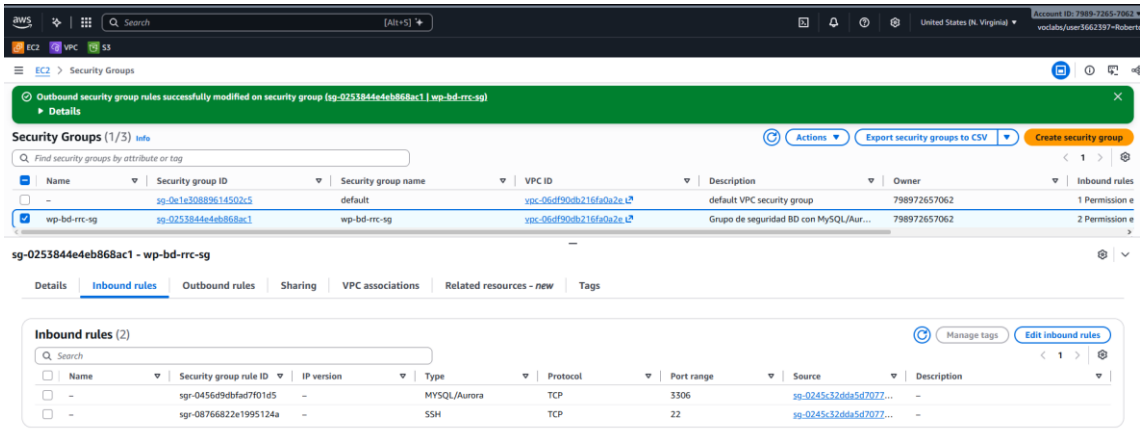


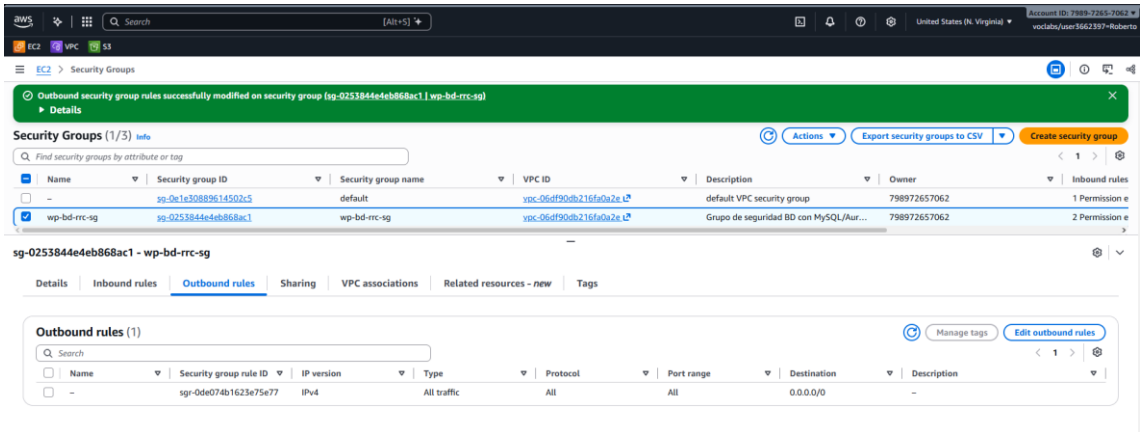
Subnet ID: subnet-0475d28eb3a9b52a4

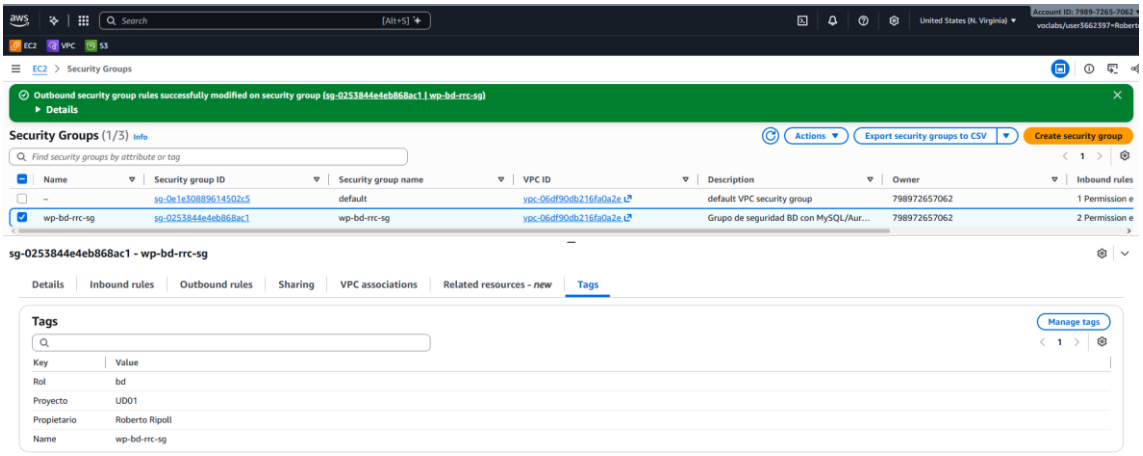
IPv4 CIDR: 172.31.32.0/20

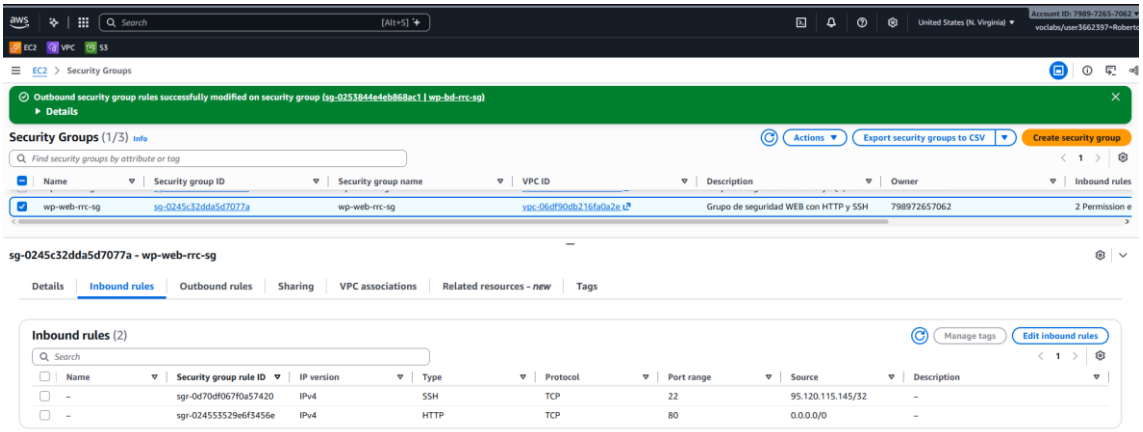
IP Personalizada: 172.31.33.10

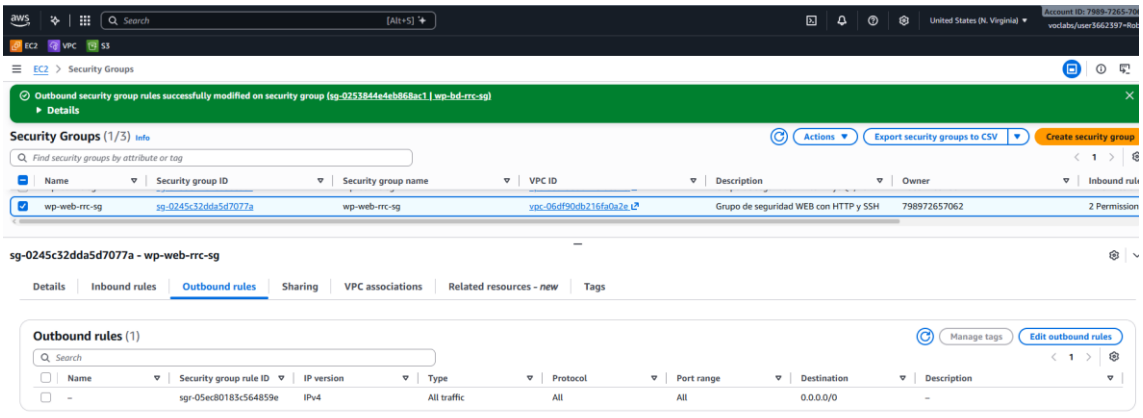
2. Grupos de seguridad

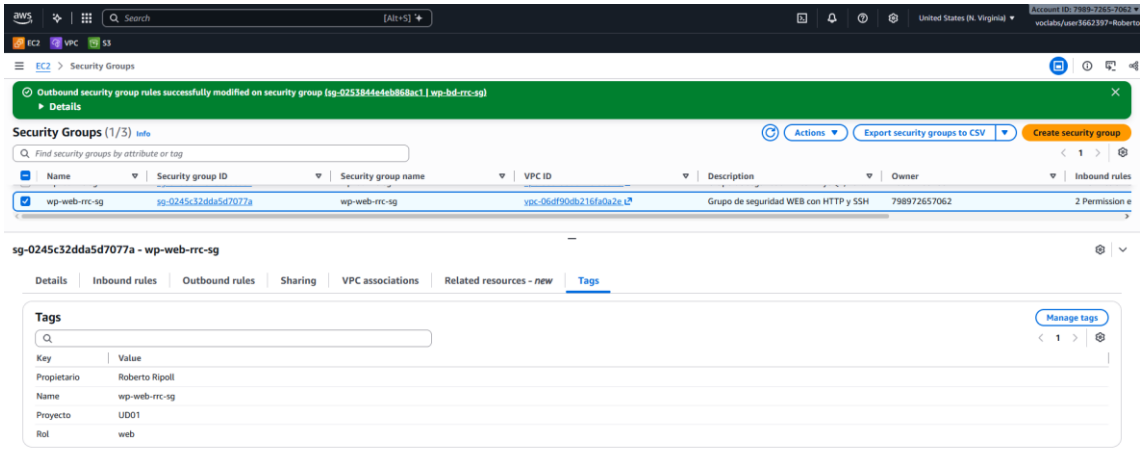




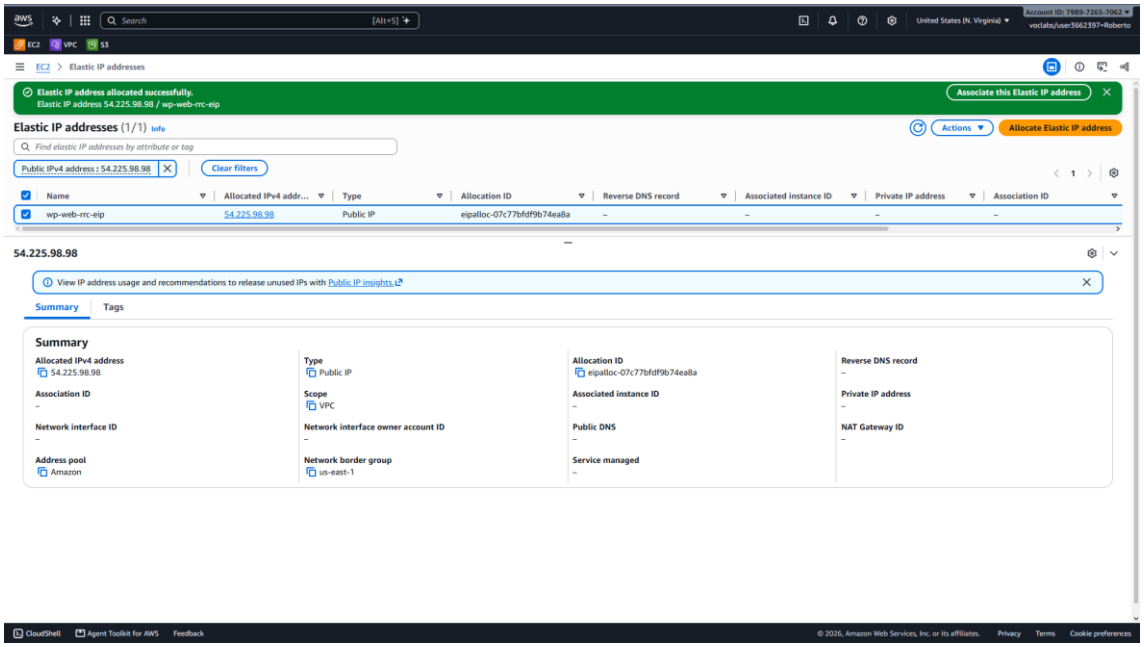

Outbound security group rules successfully modified on security group (sg-0253844e4eb868ac1 | wp-bd-rrc-sg)
Security Groups (1/3)
wp-bd-rrc-sg
sg-0253844e4eb868ac1 - wp-bd-rrc-sg
Details
Inbound rules
Outbound rules
Sharing
VPC associations
Related resources - new
Tags
Tags
Manage tags
Key
Value
Rol
bd
Proyecto
UD01
Propietario
Roberto Ripoll
Name
wp-bd-rrc-sg

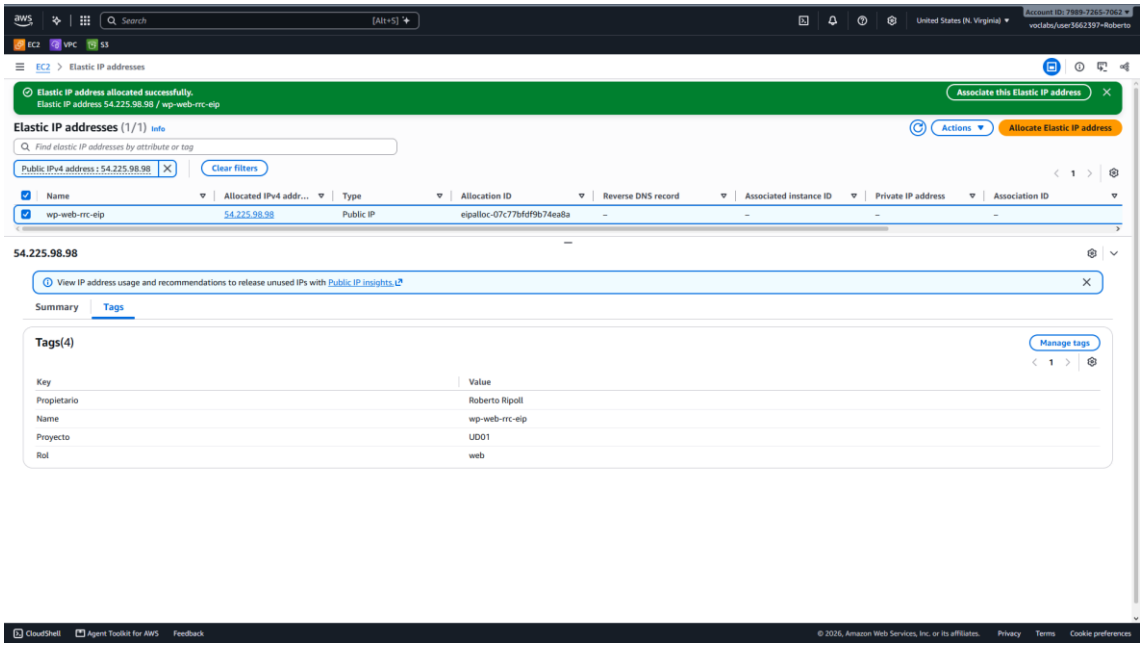

Outbound security group rules successfully modified on security group (sg-0253844e4eb868ac1 | wp-bd-rrc-sg)
Security Groups (1/3)
wp-web-rrc-sg
sg-0245c32dda5d7077a - wp-web-rrc-sg
Details
Inbound rules
Outbound rules
Sharing
VPC associations
Related resources - new
Tags
Inbound rules (2)
Manage tags
Edit inbound rules
Name
Security group rule ID
IP version
Type
Protocol
Port range
Source
Description
sgr-0d70df067f0a57420
IPv4
SSH
TCP
22
95.120.115.145/32
sgr-024553529e6f3456e
IPv4
HTTP
TCP
80
0.0.0.0/0

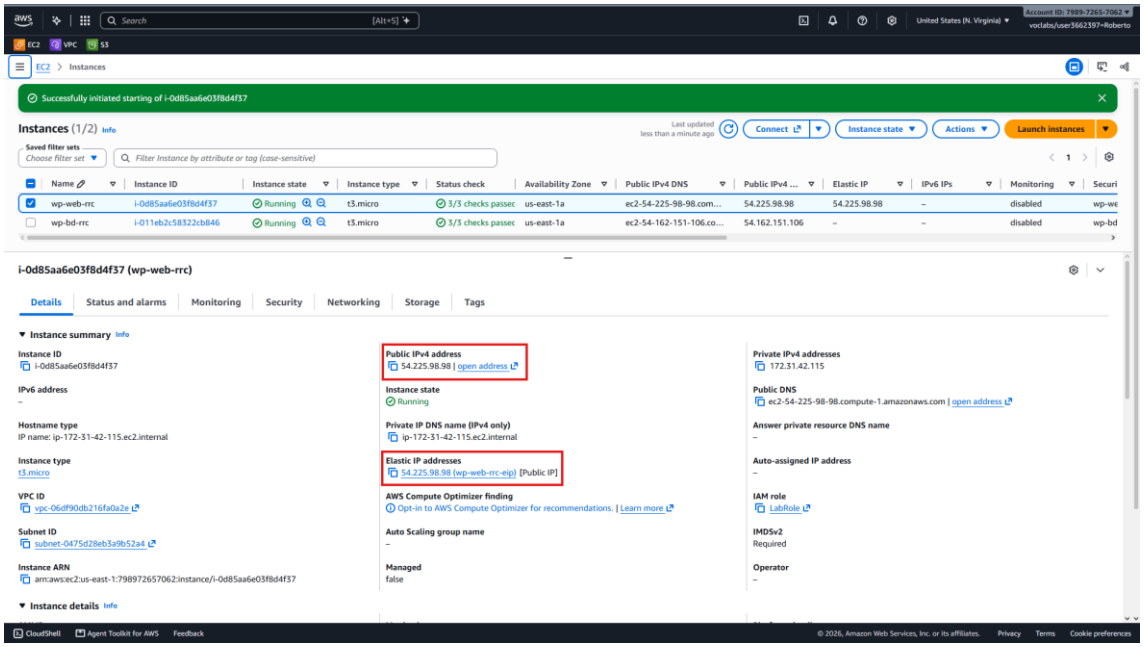

Outbound security group rules successfully modified on security group (sg-0253844e4eb868ac1 | wp-bd-rrc-sg)
Security Groups (1/3)
wp-web-rrc-sg
sg-0245c32dda5d7077a - wp-web-rrc-sg
Details
Inbound rules
Outbound rules
Sharing
VPC associations
Related resources - new
Tags
Outbound rules (1)
Manage tags
Edit outbound rules
Name
Security group rule ID
IP version
Type
Protocol
Port range
Destination
Description
sgr-05ec80183c564859e
IPv4
All traffic
All
All
0.0.0.0/0

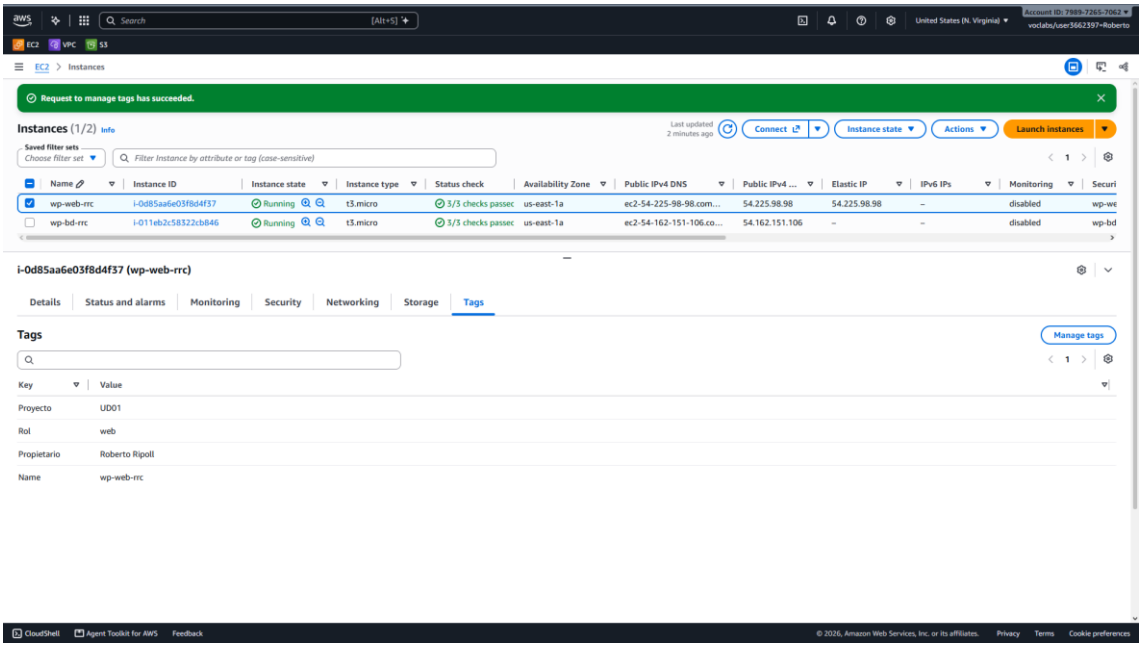


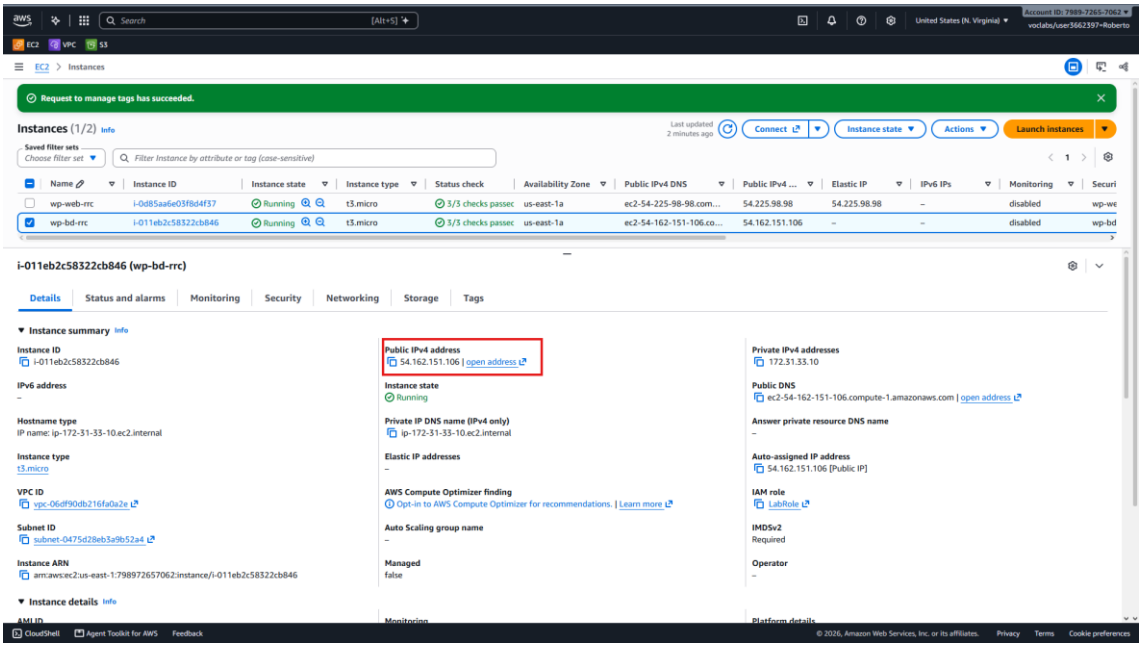
3. Instancias e IP elásticas

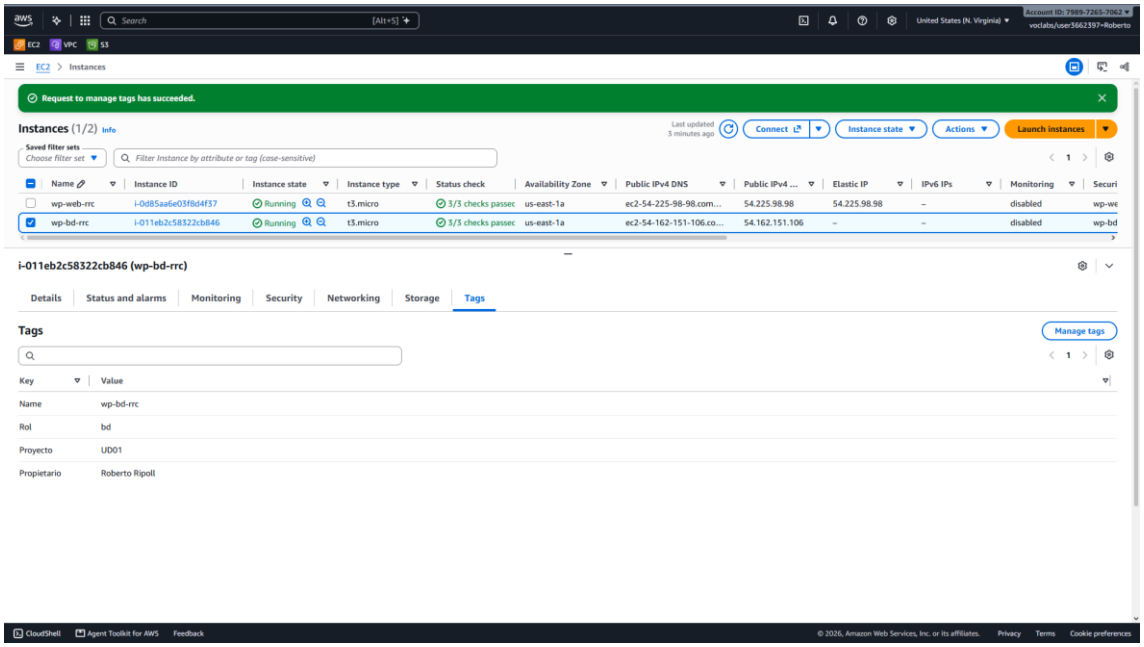



Elastic IP address allocated successfully.
Elastic IP address 54.225.98.98 / wp-web-rrc-eip
Associate this Elastic IP address
Elastic IP addresses (1/1)
Actions
Allocate Elastic IP address
Public IPv4 address : 54.225.98.98
Clear filters
Name
Allocated IPv4 addr...
Type
Allocation ID
Reverse DNS record
Associated instance ID
Private IP address
Association ID
wp-web-rrc-eip
54.225.98.98
Public IP
eipalloc-07c77bfdf9b74ea8a
54.225.98.98
View IP address usage and recommendations to release unused IPs with Public IP insights.
Summary
Tags
Tags(4)
Manage tags
Key
Value
Propietario
Roberto Ripoll
Name
wp-web-rrc-eip
Proyecto
UD01
Rol
web


Successfully initiated starting of i-0d85aa6e03f8d4f37
Instances (1/2)
Connect
Instance state
Actions
Launch instances
wp-web-rrc
i-0d85aa6e03f8d4f37
Running
t3.micro
3/3 checks passed
us-east-1a
ec2-54-225-98-98.com...
54.225.98.98
wp-bd-rrc
i-011eb2c58322cb846
ec2-54-162-151-106.co...
54.162.151.106
i-0d85aa6e03f8d4f37 (wp-web-rrc)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary
Public IPv4 address
54.225.98.98 | open address
Private IPv4 addresses
172.31.42.115
Instance state
Running
Public DNS
ec2-54-225-98-98.compute-1.amazonaws.com | open address
Private IP DNS name (IPv4 only)
ip-172-31-42-115.ec2.internal
Elastic IP addresses
54.225.98.98 (wp-web-rrc-eip) [Public IP]
Instance type
t3.micro
VPC ID
vpc-06df90db216fa0a2e
IAM role
LabRole
Subnet ID
subnet-0475d28eb3a9b52a4
Instance ARN
arn:aws:ec2:us-east-1:798972657062:instance/i-0d85aa6e03f8d4f37
Instance details


Request to manage tags has succeeded.
Instances (1/2)
Connect
Instance state
Actions
Launch instances
wp-web-rrc
i-0d85aa6e03f8d4f37
wp-bd-rrc
i-011eb2c58322cb846
i-0d85aa6e03f8d4f37 (wp-web-rrc)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Manage tags
Key
Value
Proyecto
UD01
Rol
web
Propietario
Roberto Ripoll
Name
wp-web-rrc


Request to manage tags has succeeded.
Instances (1/2)
Connect
Instance state
Actions
Launch instances
i-011eb2c58322cb846 (wp-bd-rrc)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary
Instance ID
i-011eb2c58322cb846
Public IPv4 address
54.162.151.106 | open address
Private IPv4 addresses
172.31.33.10
IPv6 address
Instance state
Running
Public DNS
ec2-54-162-151-106.compute-1.amazonaws.com | open address
Hostname type
IP name: ip-172-31-33-10.ec2.internal
Private IP DNS name (IPv4 only)
ip-172-31-33-10.ec2.internal
Answer private resource DNS name
Instance type
t3.micro
Elastic IP addresses
Auto-assigned IP address
54.162.151.106 [Public IP]
VPC ID
vpc-06df90db216fa0a2e
AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. | Learn more
IAM role
LabRole
Subnet ID
subnet-0475d28eb3a9b52a4
Auto Scaling group name
IMDSv2
Required
Instance ARN
arn:aws:ec2:us-east-1:798972657062:instance/i-011eb2c58322cb846
Managed
false
Operator
Instance details



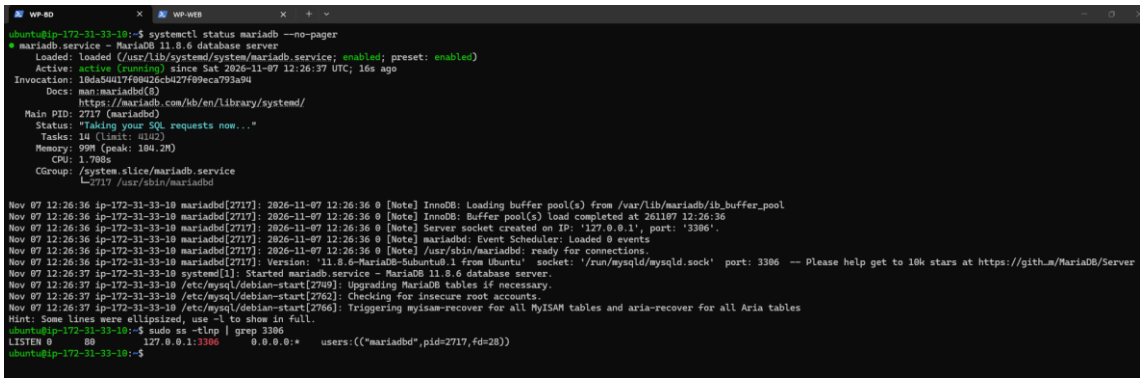
4. Acceso SSH con host de salto

```
Host wp-web
        HostName 54.225.98.98
        User ubuntu
        IdentityFile ~/.ssh/labsuser.pem

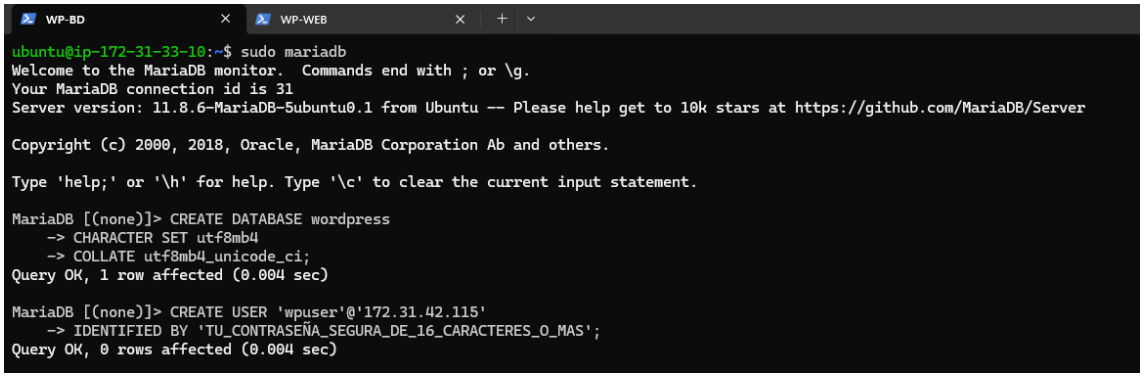
Host wp-bd
        HostName 172.31.33.10
        User ubuntu
        IdentityFile ~/.ssh/labsuser.pem
        ProxyJump wp-web
```

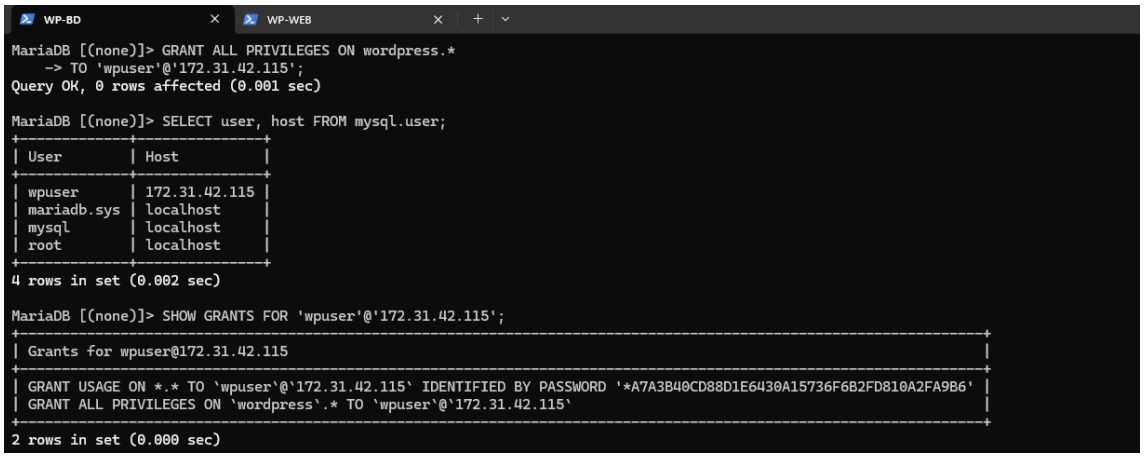
```
Windows PowerShell
PS C:\Users\canal> ssh wp-web hostname -I
172.31.42.115
PS C:\Users\canal> ssh wp-bd hostname -I
172.31.33.10
PS C:\Users\canal> ssh -o ConnectTimeout=10 -i .\.ssh\labsuser.pem ubuntu@54.162.151.106
ssh: connect to host 54.162.151.106 port 22: Connection timed out
PS C:\Users\canal>
```

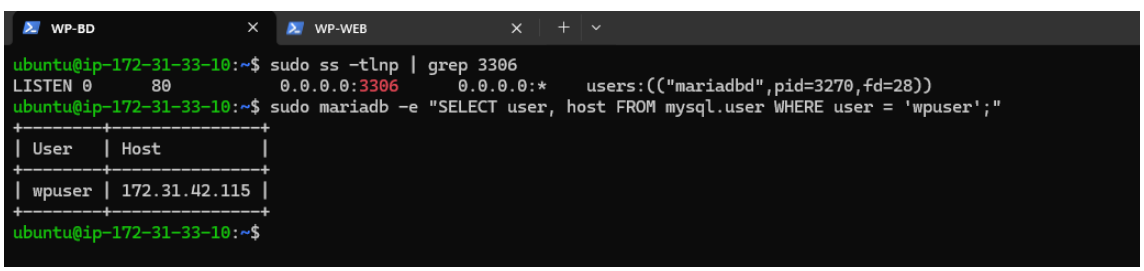
5. Capa de datos con MariaDB



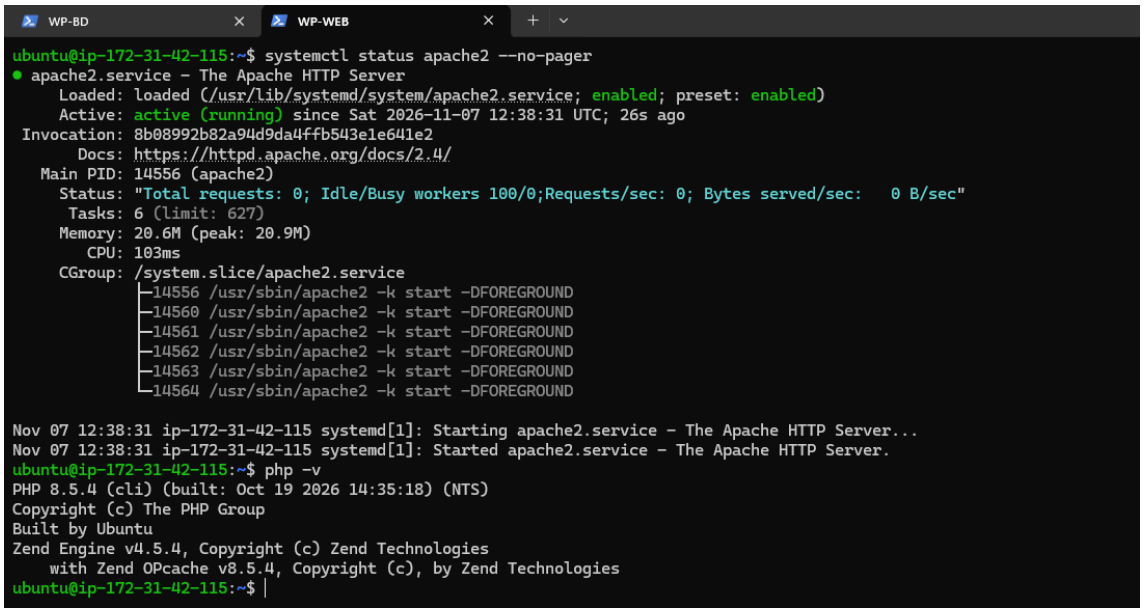

```
ubuntu@ip-172-31-33-10:~$ sudo tee /etc/mysql/mariadb.conf.d/99-wordpress.cnf > /dev/null << 'EOF'
[mysqld]
bind-address = 0.0.0.0
EOF
ubuntu@ip-172-31-33-10:~$ sudo systemctl restart mariadb
ubuntu@ip-172-31-33-10:~$ sudo ss -tlnp | grep 3306
LISTEN 0      80           0.0.0.0:3306      0.0.0.0:*    users:(("mariadbd",pid=3270,fd=28))
ubuntu@ip-172-31-33-10:~$
```

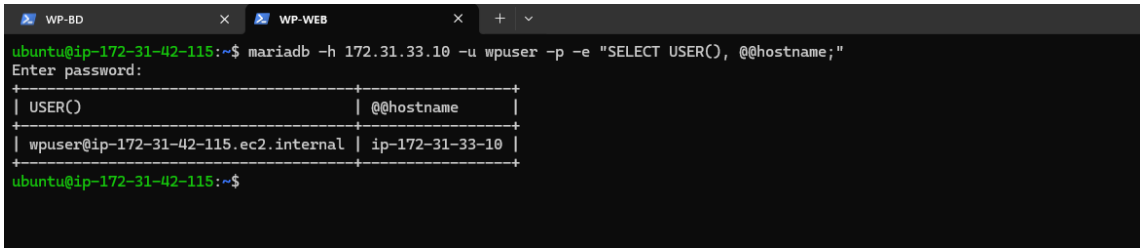


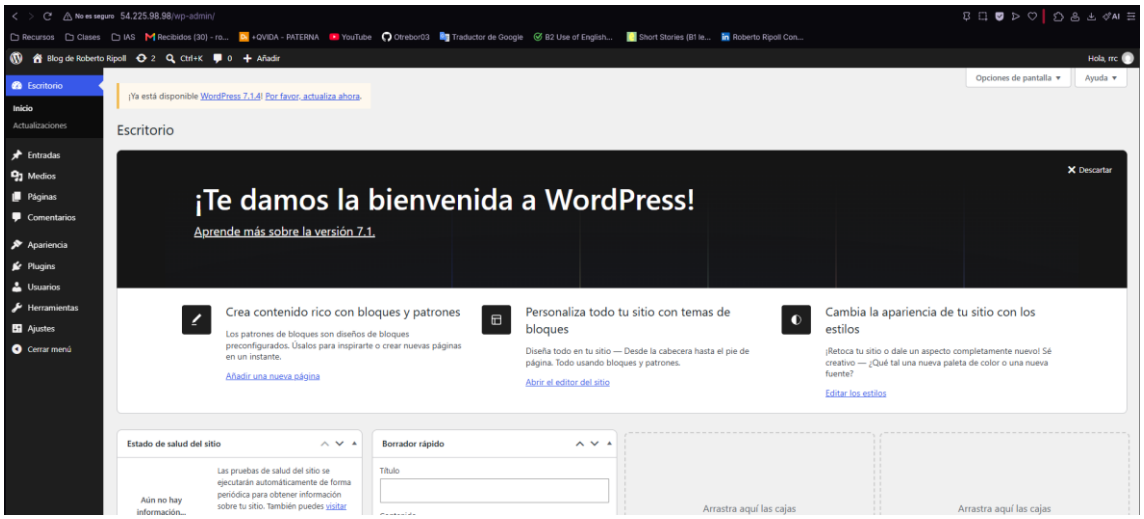




6. Capa web con Apache, PHP y Wordpress



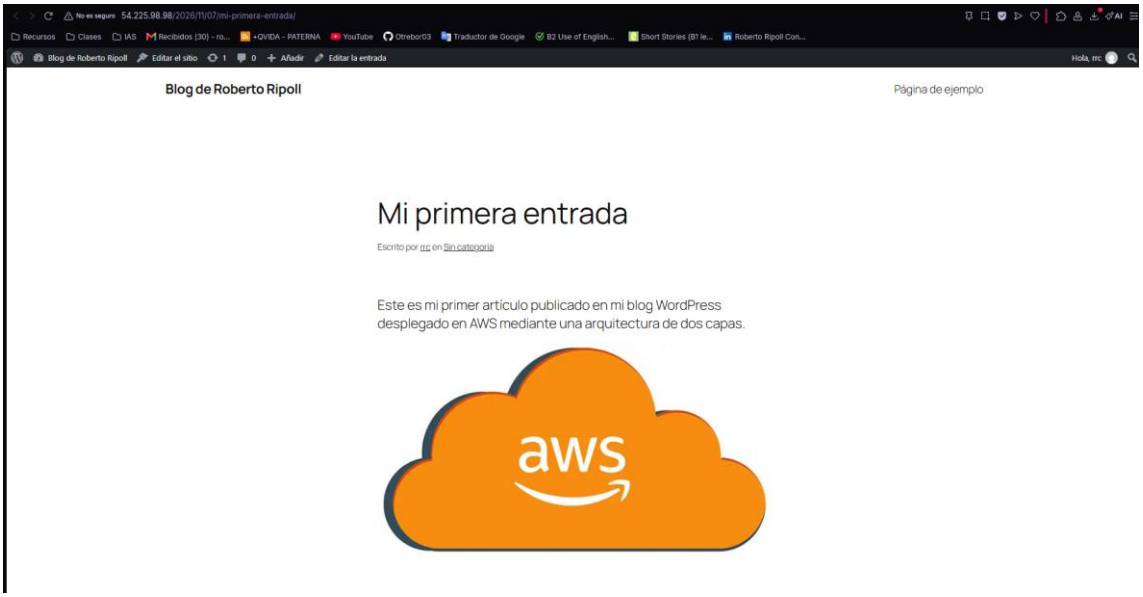




```
ubuntu@ip-172-31-42-115:~$ sudo systemctl status apache2 --no-pager
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
     Active: active (running) since Sat 2026-11-07 12:41:13 UTC; 6min ago
 Invocation: 9d8a6f8fe9dc43fd9a6870b7b71150e7
       Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 15065 (apache2)
     Status: "Total requests: 131; Idle/Busy workers 100/0;Requests/sec: 0.33; Bytes served/sec: 5.4KB/sec"
      Tasks: 11 (limit: 627)
     Memory: 103.8M (peak: 111.3M)
        CPU: 2.627s
     CGroup: /system.slice/apache2.service
             ├─15065 /usr/sbin/apache2 -k start -DFOREGROUND
             ├─15075 /usr/sbin/apache2 -k start -DFOREGROUND
             ├─15076 /usr/sbin/apache2 -k start -DFOREGROUND
             ├─15128 /usr/sbin/apache2 -k start -DFOREGROUND
             ├─15130 /usr/sbin/apache2 -k start -DFOREGROUND
             ├─15131 /usr/sbin/apache2 -k start -DFOREGROUND
             ├─15132 /usr/sbin/apache2 -k start -DFOREGROUND
             ├─15133 /usr/sbin/apache2 -k start -DFOREGROUND
             ├─15141 /usr/sbin/apache2 -k start -DFOREGROUND
             ├─15144 /usr/sbin/apache2 -k start -DFOREGROUND
             └─15154 /usr/sbin/apache2 -k start -DFOREGROUND

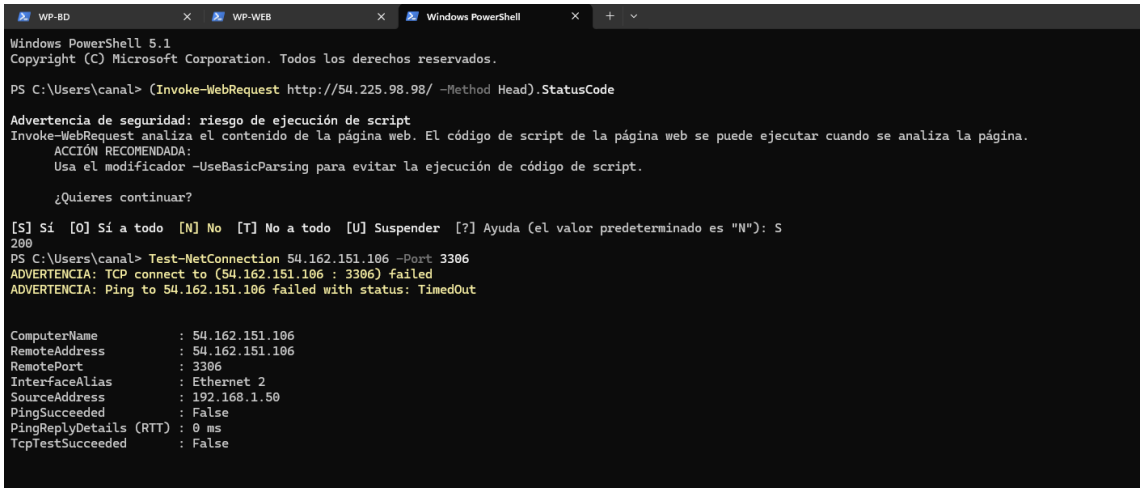
Nov 07 12:41:13 ip-172-31-42-115 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Nov 07 12:41:13 ip-172-31-42-115 systemd[1]: Started apache2.service - The Apache HTTP Server.
```

7. Primera entrada y comprobaciones

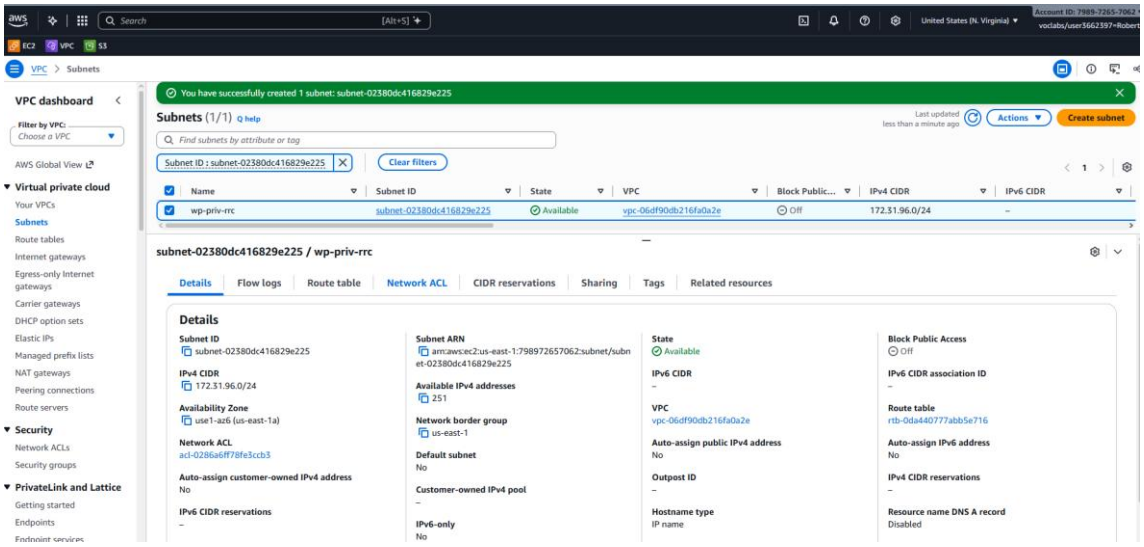


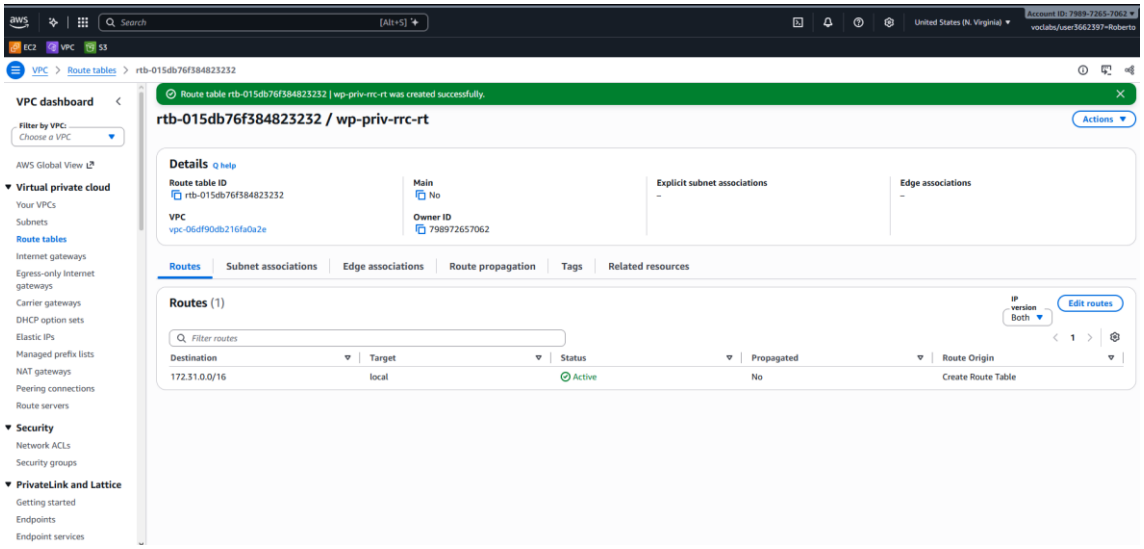

```
WP-BD | WP-WEB
ubuntu@ip-172-31-42-115:~$ find /var/www/wordpress/wp-content/uploads -type f
/var/www/wordpress/wp-content/uploads/2026/11/images-150x150.jpeg
/var/www/wordpress/wp-content/uploads/2026/11/images.jpeg
/var/www/wordpress/wp-content/uploads/2026/11/images-300x161.jpeg
ubuntu@ip-172-31-42-115:~$
```

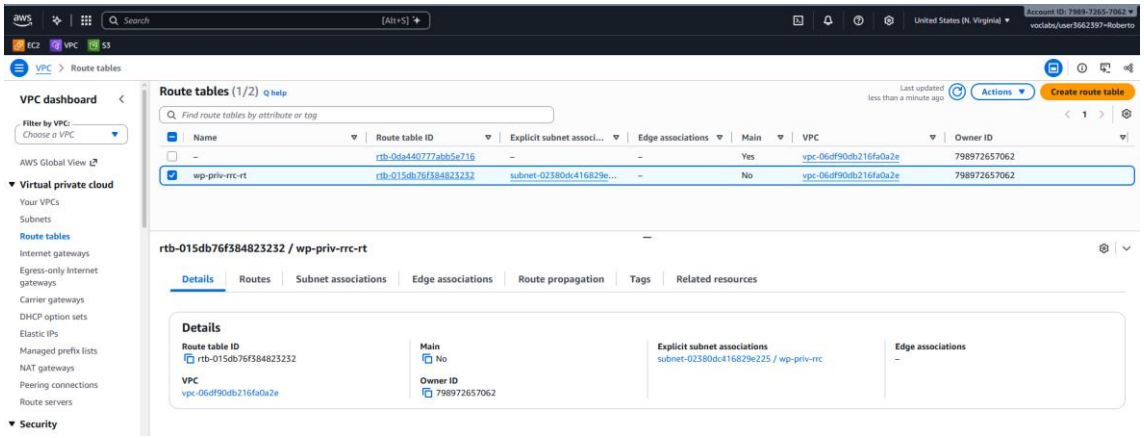
```
WP-BD | WP-WEB
ubuntu@ip-172-31-33-10:~$ sudo mariadb wordpress -e "SELECT ID, post_type, post_title FROM wp_posts WHERE post_status IN ('publish','inherit');"
+----+-----------------+--------------------+
| ID | post_type       | post_title         |
+----+-----------------+--------------------+
|  1 | post            | ¡Hola, mundo!      |
|  2 | page            | Página de ejemplo  |
|  4 | wp_navigation   | Navegación         |
|  6 | post            | Mi primera entrada |
|  7 | wp_global_styles | Custom Styles     |
|  8 | attachment      | images             |
|  9 | revision        | Mi primera entrada |
+----+-----------------+--------------------+
ubuntu@ip-172-31-33-10:~$
```

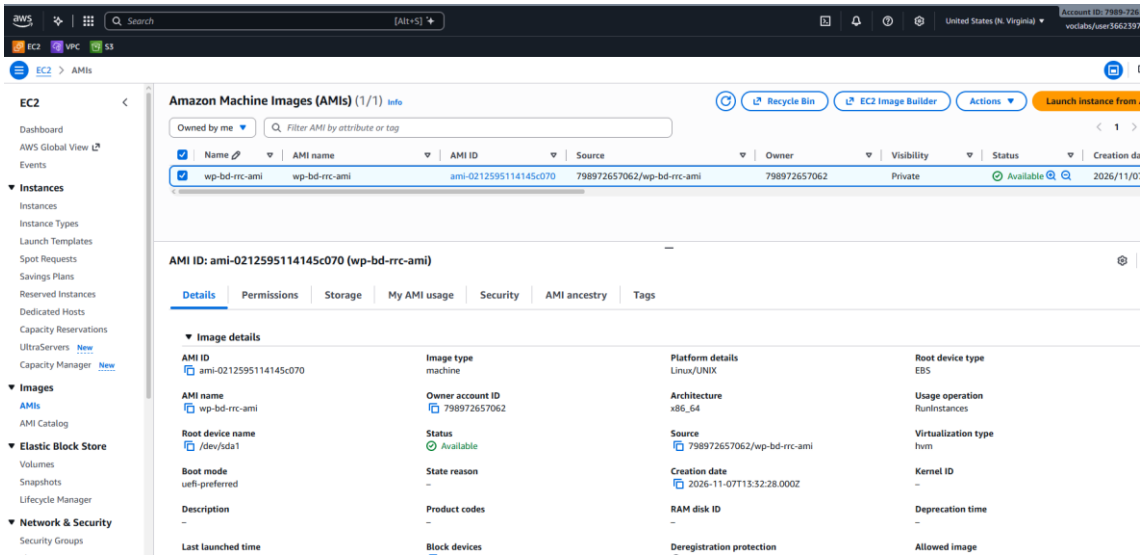


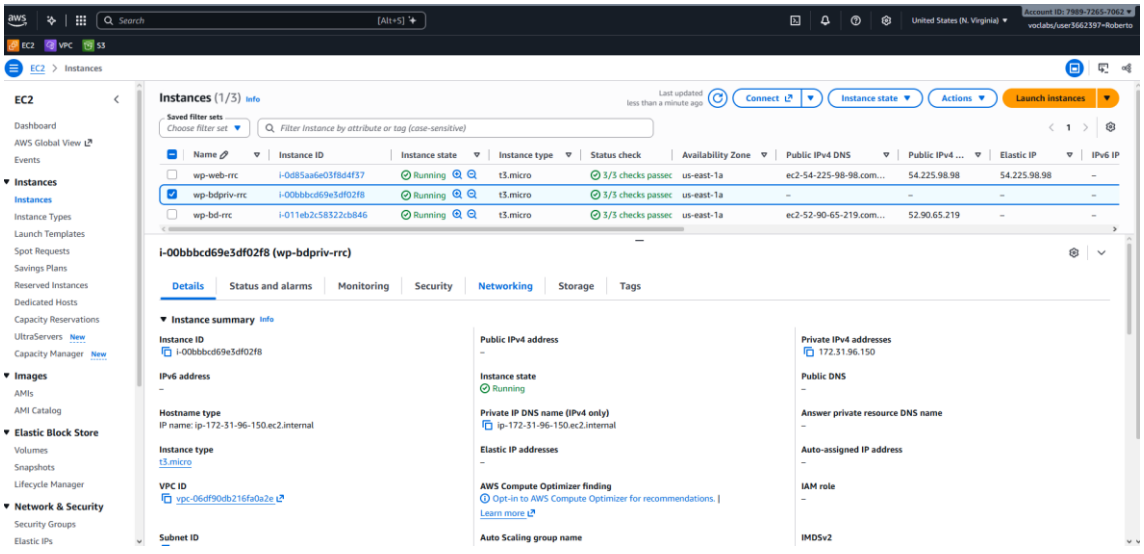
8. Base de datos en una subred privada

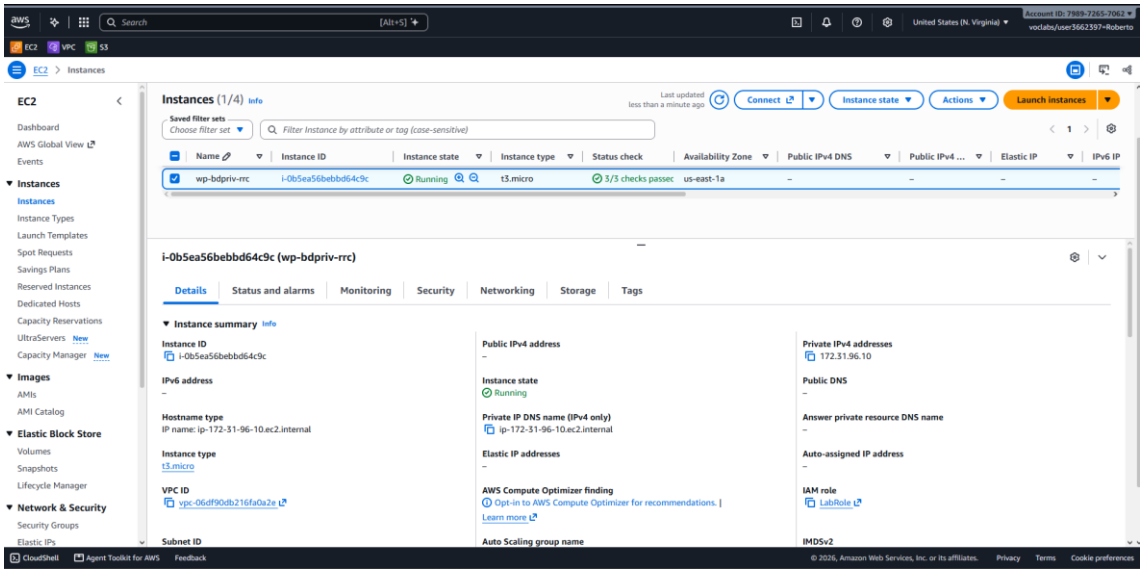





Route tables (1/2)
rtb-015db76f384823232 / wp-priv-rrc-rt
Details
Routes
Subnet associations
Edge associations
Route propagation
Tags
Related resources


Amazon Machine Images (AMIs) (1/1)
AMI ID: ami-0212595114145c070 (wp-bd-rrc-ami)
Details
Permissions
Storage
My AMI usage
Security
AMI ancestry
Tags

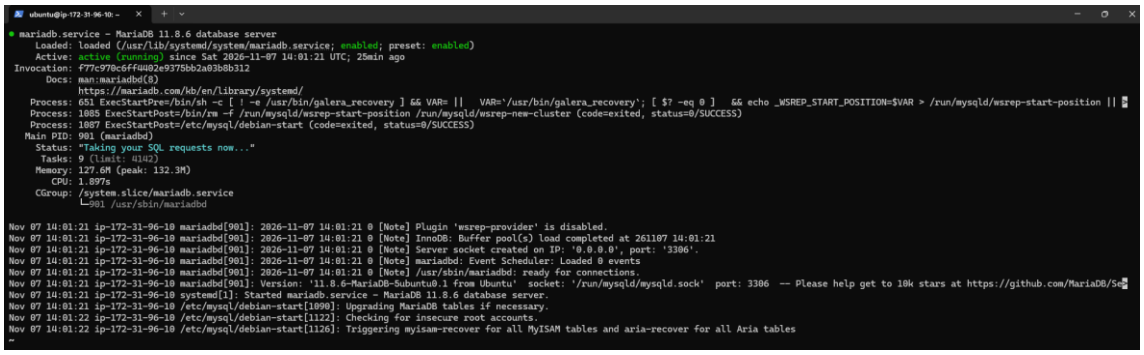

Instances (1/3)
i-00bbbcd69e3df02f8 (wp-bdpriv-rrc)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags




```
Host wp-web
        HostName 54.225.98.98
        User ubuntu
        IdentityFile ~/.ssh/labsuser.pem

Host wp-bd
        HostName 172.31.96.10
        User ubuntu
        IdentityFile ~/.ssh/labsuser.pem
        ProxyJump wp-web
```





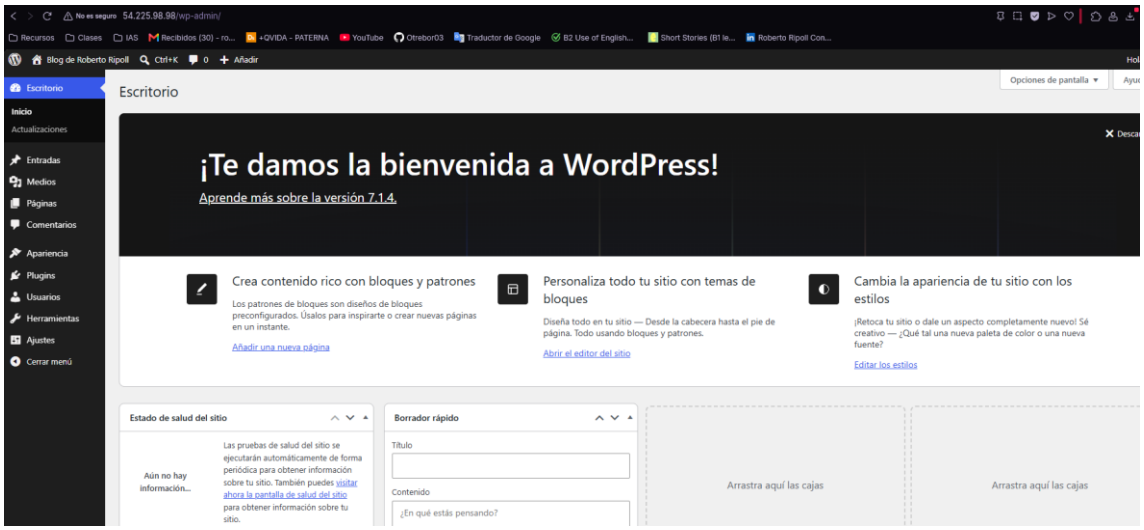
```
ubuntu@ip-172-31-96-10: ~
ubuntu@ip-172-31-96-10:~$ sudo ss -lntp | grep 3306
LISTEN 0      80           0.0.0.0:3306       0.0.0.0:*    users:(("mariadbd",pid=901,fd=40))
ubuntu@ip-172-31-96-10:~$
```

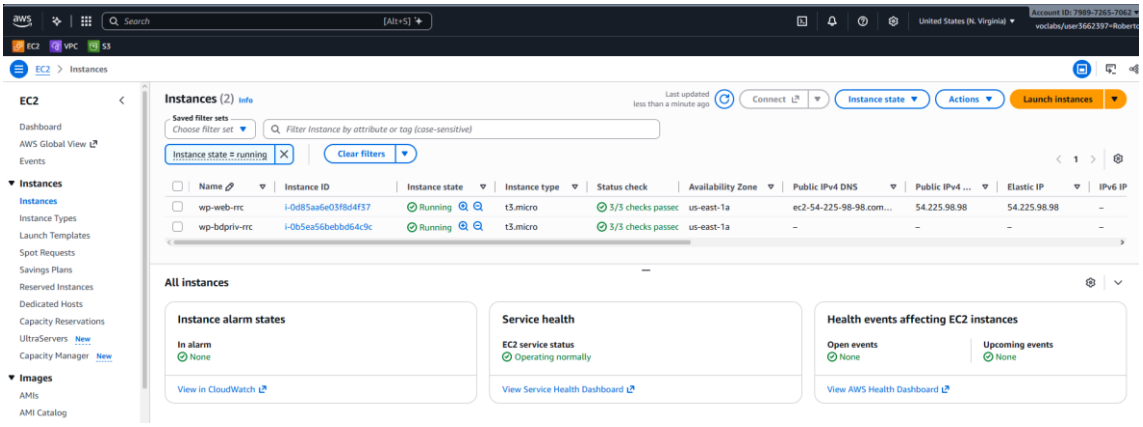
```
WP-DB | WP-WEB
ubuntu@ip-172-31-42-115:~$ mariadb -h 172.31.96.10 -u wpuser -p -e "SELECT USER(), @@hostname;"
Enter password:
+------------------------------------+-----------------+
| USER()                             | @@hostname      |
+------------------------------------+-----------------+
| wpuser@ip-172-31-42-115.ec2.internal | ip-172-31-96-10 |
+------------------------------------+-----------------+
ubuntu@ip-172-31-42-115:~$
```

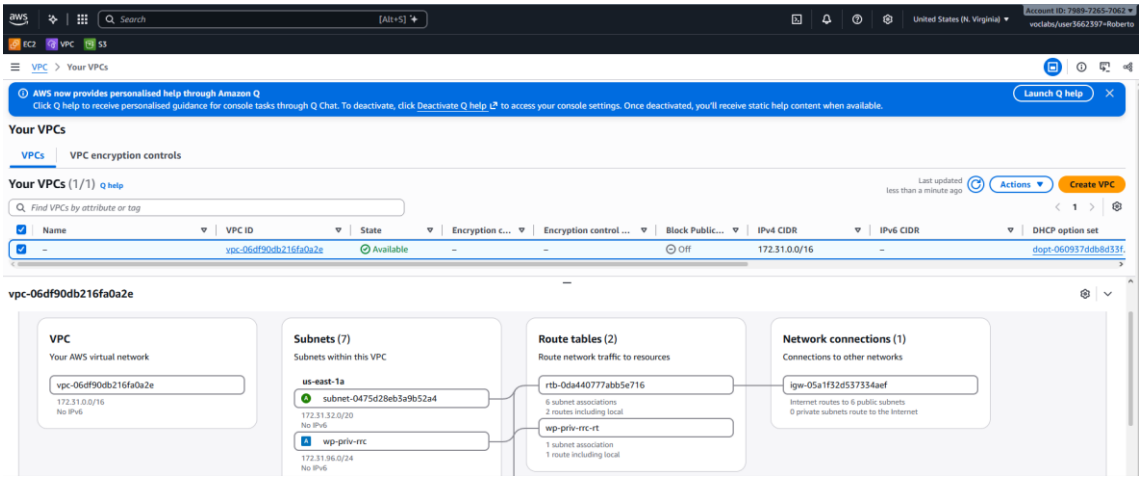
```
WP-DB | WP-WEB
ubuntu@ip-172-31-42-115:~$
ubuntu@ip-172-31-42-115:~$ sudo grep DB_HOST /var/www/wordpress/wp-config.php
define( 'DB_HOST', '172.31.33.10' );
ubuntu@ip-172-31-42-115:~$ sudo sed -i "s/172\.31\.33\.10/172.31.96.10/g" /var/www/wordpress/wp-config.php
ubuntu@ip-172-31-42-115:~$ sudo grep DB_HOST /var/www/wordpress/wp-config.php
define( 'DB_HOST', '172.31.96.10' );
ubuntu@ip-172-31-42-115:~$
```

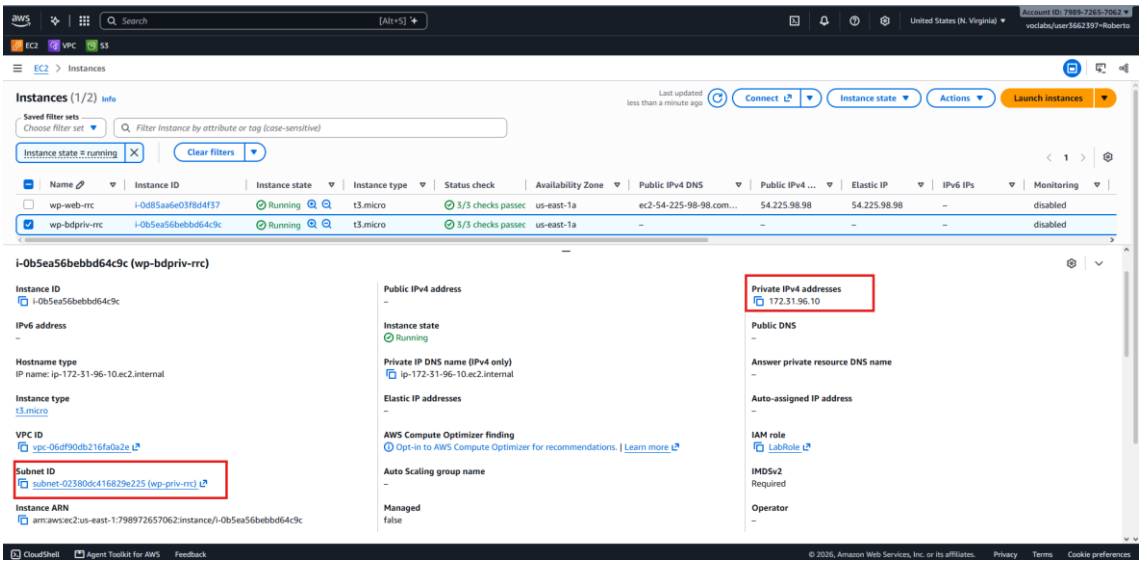
```
WP-DB | WP-WEB | Windows PowerShell
Windows PowerShell 5.1
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PS C:\Users\canal> (Invoke-WebRequest "http://54.225.98.98/" -UseBasicParsing).StatusCode
200
PS C:\Users\canal>
```




EC2 > Instances
Instances (2)
wp-web-rrc
i-0d85aa6e03f8d4f37
wp-bdpriv-rrc
i-0b5ea56bebbd64c9c
All instances
Instance alarm states
Service health
Health events affecting EC2 instances


VPC > Your VPCs
Your VPCs
vpc-06df90db216fa0a2e
Subnets (7)
Route tables (2)
Network connections (1)
subnet-0475d28eb3a9b52a4
wp-priv-rrc
rtb-0da440777abb5e716
igw-05a1f32d537334aef


Instances (1/2)
wp-web-rrc
wp-bdpriv-rrc
i-0b5ea56bebbd64c9c (wp-bdpriv-rrc)
Private IPv4 addresses
172.31.96.10
Subnet ID
subnet-02380dc416829e225 (wp-priv-rrc)

9. Para pensar

9.1. ¿Qué ventaja tiene usar como origen el grupo wp-web-xx-sg en lugar de la IP privada de wp-web-xx?

Al emplear el grupo de seguridad como origen permite autorizar las conexiones de cualquier instancia que pertenezca a wp-web-rrc-sg sin necesidad de depender de una dirección IP concreta. En caso de sustituir el servidor o añadir una segunda instancia con el mismo grupo de seguridad, podrá conectarse a MariaDB sin modificar las reglas de entrada. Esto facilita el mantenimiento y escalabilidad de la arquitectura.

9.2. wp-bd-xx tiene IP pública, pero nadie puede conectarse a ella desde Internet. ¿Qué elemento lo impide? ¿Por qué no basta con quitarle la IP pública para que siga funcionando todo igual?

Lo impide el grupo de seguridad wp-bd-rrc-sg que únicamente permite conexiones SSH por el puerto 22 y MariaDB por el puerto 3306. La IP publica no implica que los puertos estén abiertos a internet. Si quitamos esa IP la instancia seguirá comunicándose con el servidor mediante la IP privada pero será imposibilitara la descarga de paquetes/actualizaciones.